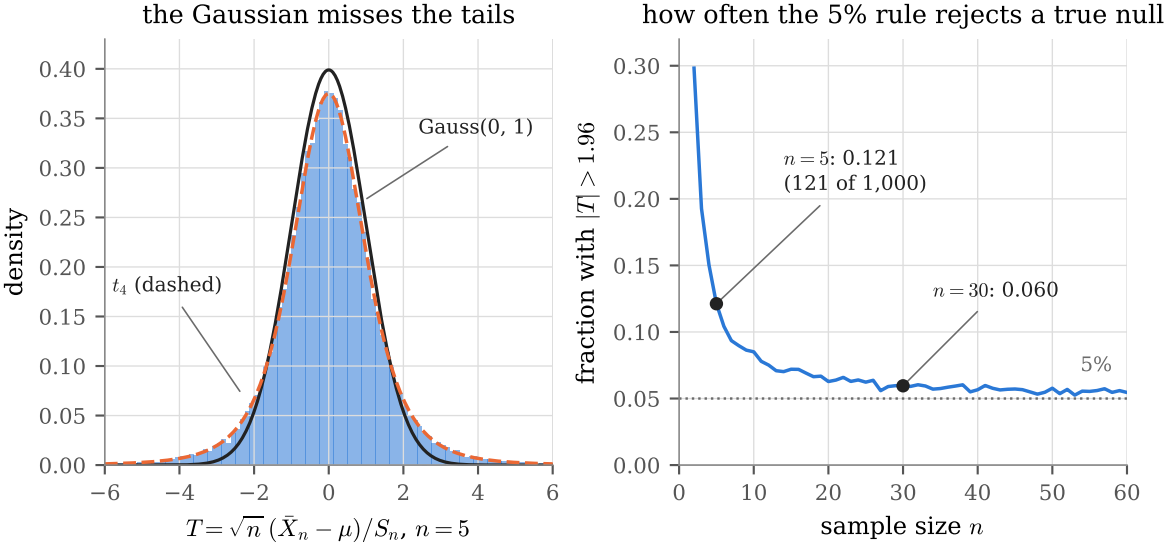
the Gaussian misses the tails
density
Gauss(0, 1)
t_4 (dashed)
$T = \sqrt{n}\,(\bar{X}_n - \mu)/S_n$, $n = 5$
how often the 5% rule rejects a true null
fraction with $|T| > 1.96$
$n = 5$: 0.121
(121 of 1,000)
$n = 30$: 0.060
5%
sample size n